APEXPLANET

CYBERSECURITY & ETHICAL HACKING

TASK 1 REPORT

Foundations of Cybersecurity
& Environment Setup

Candidate detail	Information
Name	Devaraju GR
Internship domain	Cybersecurity & Ethical Hacking
Organization	ApexPlanet Software Pvt. Ltd.
Task	Task 1 – Foundations & Environment Setup
Timeline in task brief	Days 1–12
Report status	Starter report – practical evidence pending

Prepared as a working report template and practical guide. Update the evidence and completion fields after carrying out each activity in your own authorized lab.

Ethical use only • Isolated lab testing • Document genuine results

Report status and instructions

Important: You indicated that Task 1 has not been started. Therefore, this PDF is a structured report and execution guide, not proof that the practical work is complete. Do not submit it as a completed practical report until you have performed the activities, replaced the evidence placeholders with your own screenshots, and recorded your actual results.

How to use this report

• Work through the sections in order and use only systems you own or have explicit permission to test.

• Capture genuine screenshots with the relevant application window and a short explanatory caption.

• Record your machine details, tool versions, commands, outputs, dates and any issues encountered.

• Replace every [ADD YOUR EVIDENCE] marker with your own result before creating the final submission PDF.

• Keep vulnerable targets such as Metasploitable or DVWA on an isolated Host-Only network; never expose them to the public internet.

Task 1 objectives

Build foundational knowledge of cybersecurity, understand common threats, create a safe penetration-testing lab, practice Linux and networking fundamentals, learn introductory cryptography, and become familiar with essential security tools.

Expected deliverables from the task brief

Deliverable	Evidence to include
Lab setup report	Screenshots of Kali Linux, the vulnerable target environment and a Wireshark test capture.
GitHub repository	Task notes, practical documentation, Linux command notes and a cheat sheet.
5-minute video	A screen-recorded walkthrough of the lab setup and work completed, uploaded to LinkedIn.

Contents

Section	Topic
1	Cybersecurity foundations and CIA Triad
2	Threat types and attack vectors
3	Lab environment setup
4	Linux fundamentals
5	Networking basics
6	Cryptography basics and OpenSSL
7	Security tool familiarization
8	Evidence checklist, GitHub structure and video plan
9	Glossary and references

1. Cybersecurity foundations

Cybersecurity is the practice of protecting computers, networks, applications and data from unauthorized access, misuse, disruption or damage. Ethical hacking uses authorized security testing to identify weaknesses so they can be fixed before malicious actors exploit them.

1.1 The CIA Triad

Principle	Meaning	Example
Confidentiality	Only authorized people can access information.	Access controls and encryption protect private files.
Integrity	Information remains accurate and is changed only by authorized actions.	Hashes and audit logs help detect unauthorized changes.
Availability	Systems and data are accessible when needed.	Backups, redundancy and DDoS protection support availability.

1.2 Security objectives

• Identify assets such as devices, applications, accounts and sensitive information.

• Assess threats, vulnerabilities, likelihood and potential impact.

• Apply safeguards such as authentication, patching, encryption, backups and monitoring.

• Detect suspicious activity and respond using a documented incident process.

2. Common threats and attack vectors

Threat	Description	Basic defensive measure
Phishing	Deceptive messages or websites try to steal credentials or deliver malware.	Verify senders and URLs; use MFA and awareness training.
Malware	Software designed to damage, spy on or gain unauthorized access to systems.	Patch systems, use reputable endpoint protection and avoid unknown downloads.
DDoS	Many sources overwhelm a service and reduce its availability.	Use rate limiting, traffic filtering and provider-level DDoS protection.
SQL injection	Untrusted input is interpreted as part of a database query.	Use parameterized queries, input validation and least-privilege DB accounts.
Brute force	Repeated password guesses attempt to gain account access.	Use MFA, rate limits, strong unique passwords and lockout controls.
Ransomware	Malware encrypts or otherwise blocks access to data and demands payment.	Maintain offline/immutable backups, patch and rehearse recovery.

2.1 Social engineering, wireless and insider threats

• **Social engineering:** Manipulating people into revealing information or taking unsafe actions. Verify unusual requests through a trusted second channel.

• **Wireless threats:** Risks include rogue access points, weak Wi-Fi passwords and insecure protocols. Use WPA2-AES or WPA3, update router firmware and disable insecure legacy settings.

• **Insider threats:** Risk can come from malicious, careless or compromised insiders. Apply least privilege, logging, separation of duties and timely access reviews.

2.2 Responsible testing boundary

Only scan or test systems that you own or are explicitly authorized to assess. Do not test public IP addresses, college networks, employer systems or third-party websites without written permission. Use the isolated practice environment described below.

3. Lab environment setup

A controlled lab allows security concepts and tools to be practiced without affecting real users or production systems. The task brief allows VirtualBox or VMware, Kali Linux as the testing machine, and Metasploitable 2 or DVWA as a deliberately vulnerable target.

3.1 Suggested lab topology

Machine	Role	Network setting
Host computer	Runs the virtualization software.	Physical network; do not bridge the vulnerable target.
Kali Linux VM	Security testing workstation.	Host-Only adapter for lab traffic; optional NAT only when needed for updates.
Metasploitable 2 / DVWA VM	Intentionally vulnerable target for learning.	Host-Only adapter only; no bridged or public exposure.

3.2 Setup procedure

• Install Oracle VirtualBox or VMware Workstation from its official website.

• Create or import a Kali Linux virtual machine using the official Kali image. Allocate resources appropriate to your computer.

• Create a Host-Only virtual network in the virtualization manager.

• Attach the Kali VM to the Host-Only network. If internet access is required for updates, temporarily use a separate NAT adapter and avoid routing the vulnerable target to the internet.

• Import or install Metasploitable 2 or deploy DVWA in a separate target VM/container connected only to the Host-Only network.

• Start both systems and verify that they can communicate on the isolated lab network.

• Take a snapshot before testing so the target can be restored.

3.3 Verification commands

```
# On Kali: identify network interfaces and IP configuration
ip addr
ip route

# Test connectivity to your own lab target only
ping -c 4 <LAB_TARGET_IP>

# Show listening services on your own Kali machine
ss -tuln
```

Replace <LAB_TARGET_IP> with the address assigned to your own vulnerable target VM. Do not copy an address from an unrelated network.

3.4 Evidence to capture

Evidence ID	Screenshot / proof	Status
LAB-01	VirtualBox/VMware showing Kali VM configuration.	[ADD YOUR EVIDENCE]
LAB-02	Kali Linux desktop and terminal showing ip addr.	[ADD YOUR EVIDENCE]
LAB-03	Vulnerable target VM running in the isolated lab.	[ADD YOUR EVIDENCE]
LAB-04	Host-Only network configuration and connectivity test.	[ADD YOUR EVIDENCE]

4. Linux fundamentals

Kali Linux is based on Debian and provides a Linux environment commonly used for security education and authorized testing. The following commands are introductory administration and navigation commands.

Command	Purpose	Example
pwd	Print current directory.	pwd
ls -la	List files, including hidden files and details.	ls -la
cd	Change directory.	cd ~/Documents
mkdir	Create a directory.	mkdir task1-notes
touch	Create an empty file.	touch notes.txt
cp / mv	Copy or move files.	cp notes.txt notes-copy.txt
cat / less	Display file contents.	less notes.txt
grep	Search text patterns.	grep -n 'TODO' notes.txt
find	Find files by name or criteria.	find . -name '*.txt'
chmod	Change file permissions.	chmod 640 notes.txt
chown	Change file ownership (admin use).	sudo chown user:user notes.txt
apt update	Refresh package metadata.	sudo apt update
apt upgrade	Upgrade installed packages.	sudo apt upgrade
ip addr	Show IP addresses and interfaces.	ip addr
ip route	Show routing table.	ip route
ss -tuln	Show listening TCP/UDP sockets.	ss -tuln

4.1 File permissions

Linux permissions are commonly represented by read (r), write (w) and execute (x) for the owner, group and others. Numeric values are r=4, w=2 and x=1. For example, 640 means owner read/write, group read, and others no access.

```
# View permissions
ls -l notes.txt

# Set owner read/write, group read, no permissions for others
chmod 640 notes.txt

# View current user and groups
whoami
id
```

4.2 Package management

```
# Refresh package lists
sudo apt update

# Upgrade packages (review the changes before confirming)
sudo apt upgrade

# Search for a package
apt search wireshark

# Install a package from the configured repository
sudo apt install wireshark
```

Use sudo only when needed. Review package sources and prompts before installing or upgrading software.

4.3 Linux evidence

[ADD YOUR EVIDENCE: screenshot of terminal showing pwd, ls -la, permissions using ls -l, and network information using ip addr. Add a short caption describing what each command demonstrates.]

5. Networking basics

Networking allows devices to exchange data. Security professionals need to understand how traffic is addressed, routed and carried by common protocols.

5.1 OSI model

Layer	Name	Examples / role
7	Application	HTTP, HTTPS, DNS; services used by applications.
6	Presentation	Data representation, encoding and encryption concepts.
5	Session	Establishes, manages and ends communication sessions.
4	Transport	TCP and UDP; ports and end-to-end transport.
3	Network	IP addressing and routing.
2	Data Link	Frames, MAC addresses and local network delivery.
1	Physical	Signals, cables, radio and physical transmission.

5.2 TCP/IP model

TCP/IP layer	Typical protocols / purpose
Application	HTTP/HTTPS, DNS, SSH, SMTP; application services.
Transport	TCP (reliable connection-oriented delivery), UDP (connectionless datagrams).
Internet	IPv4/IPv6 and routing between networks.
Network Access	Ethernet, Wi-Fi and local-link transmission.

5.3 Key concepts

• **IP address:** A logical address used to identify an interface on an IP network.

• **Subnet mask / prefix:** Defines which portion of an IP address identifies the network.

• **Default gateway:** Router used to reach destinations outside the local subnet.

• **DNS:** Resolves domain names to IP addresses and other records.

• **HTTP / HTTPS:** Web application protocols; HTTPS protects HTTP traffic using TLS.

• **NAT:** Translates network addresses, commonly allowing private hosts to share a public IPv4 address.

5.4 Useful network commands

```
# Interface addresses and routes
ip addr
ip route

# DNS lookup
nslookup example.com
# or
dig example.com

# Show route to a destination
traceroute example.com

# Test name resolution / reachability
ping -c 4 example.com
```

Run these commands on your own computer or lab. Network reachability depends on firewall settings and the environment; a failed ping does not necessarily mean a host is offline.

5.5 Networking evidence

[ADD YOUR EVIDENCE: terminal screenshots showing interface address, default route and a DNS lookup. Redact public IPs or personal network details before publishing.]

6. Cryptography basics

Cryptography protects information using mathematical techniques. It supports confidentiality, integrity, authentication and secure communication.

Concept	How it works	Common use
Symmetric encryption	The same secret key encrypts and decrypts data.	Efficient encryption of files and data streams; e.g. AES.
Asymmetric cryptography	A public/private key pair supports encryption, key exchange or digital signatures.	TLS certificates and digital signatures; e.g. RSA or elliptic-curve cryptography.
Hashing	A one-way digest is calculated from data; it is not decryption.	Integrity checks and password storage (with dedicated password-hashing schemes).
Digital certificate	Binds an identity to a public key and is signed by a trusted certificate authority.	Authenticating websites and services.
TLS	A protocol that establishes an authenticated, encrypted channel.	HTTPS and other secure network services.

6.1 Hash a file with OpenSSL

```
# Create a small test file
printf 'ApexPlanet Task 1 practice' > sample.txt

# Calculate SHA-256 digest
openssl dgst -sha256 sample.txt

# Change the file and calculate again
printf ' - updated' >> sample.txt
openssl dgst -sha256 sample.txt
```

The digest should change when the file content changes. A hash is useful for integrity checking; it does not encrypt or conceal the file.

6.2 Generate a local test key and self-signed certificate

```
# Generate a private RSA key for a local learning exercise
openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:2048 -out lab-private.key

# Create a self-signed certificate for local testing
openssl req -new -x509 -key lab-private.key -out lab-certificate.crt -days 30

# Inspect the certificate
openssl x509 -in lab-certificate.crt -text -noout
```

Use test keys only. Protect private keys and never upload them to GitHub. A self-signed certificate is suitable for a local demonstration but is not automatically trusted by browsers or other systems.

6.3 Cryptography evidence

[ADD YOUR EVIDENCE: SHA-256 command output and certificate details. Do not include private key contents or secrets in screenshots or the repository.]

7. Security tool familiarization

Tool	Purpose	Task 1 learning activity
Wireshark	Captures and inspects network packets.	Capture traffic generated by your own lab machine and identify DNS, TCP or HTTP/TLS metadata.
Nmap	Discovers hosts and services on authorized networks.	Scan only your isolated lab target and document the discovered ports and services.
Burp Suite	Intercepts and analyzes web application requests.	Use the browser and a deliberately vulnerable local DVWA instance; inspect your own test traffic.
Netcat	A networking utility for testing TCP/UDP connections and simple data transfer.	Demonstrate a local connection between two lab terminals, then explain the client/server roles.

7.1 Wireshark – safe starter workflow

• Open Wireshark and select the interface associated with your isolated lab network.

• Start a capture, then generate a small amount of traffic from your own lab machine, such as a DNS lookup.

• Stop the capture and inspect the packet list.

• Try display filters such as dns, tcp, or tls. Filters narrow the displayed packets; they do not change the original capture.

• Save a small capture file only if it contains no sensitive information. Review it before sharing.

7.2 Nmap – authorized lab target only

```
# Discover common TCP ports on your own lab target
nmap --top-ports 100 <LAB_TARGET_IP>

# Identify service/version information on the lab target
nmap -sV <LAB_TARGET_IP>
```

Do not run these scans against public systems or networks without explicit authorization. Replace the placeholder only with the IP of your isolated practice target.

7.3 Burp Suite – local web application

• Launch Burp Suite and use its built-in browser or configure a dedicated test browser.

• Open only your local DVWA test application.

• Enable interception and observe a request generated by your own browser.

• Forward or drop the request and observe the browser response.

• Do not use captured credentials or test requests against real services.

7.4 Netcat – local connection demonstration

```
# Terminal 1: listen on a local high port
nc -l -p 4444

# Terminal 2: connect to the same machine
nc 127.0.0.1 4444

# Type a short test message in either terminal.
```

Option syntax can differ between Netcat versions. Check nc -h if your installed version uses a different listener syntax. Stop the test with Ctrl+C.

7.5 Tool evidence checklist

Evidence ID	Required screenshot / result	Status
TOOL-01	Wireshark capture showing packets from your lab activity.	Pending
TOOL-02	Nmap output for your isolated target, with target identity visible.	Pending
TOOL-03	Burp Suite showing a request to your local DVWA application.	Pending
TOOL-04	Netcat local connection and test message.	Pending

8. Final evidence, GitHub and video submission

8.1 Screenshot naming convention

```
Task-1/
  screenshots/
    01-kali-desktop.png
    02-virtual-machine-settings.png
    03-host-only-network.png
    04-target-vm-running.png
    05-linux-commands.png
    06-networking-commands.png
    07-openssl-sha256.png
    08-openssl-certificate.png
    09-wireshark-capture.png
    10-nmap-lab-scan.png
    11-burp-local-dvwa.png
    12-netcat-local-test.png
```

Use screenshots from your own environment. Avoid including passwords, private keys, personal messages, unrelated browser tabs or sensitive network information.

8.2 Suggested GitHub repository structure

```
ApexPlanet-Cybersecurity-Internship/
■■■ README.md
■■■ Task-1/
■   ■■■ README.md
■   ■■■ notes/
■   ■   ■■■ cybersecurity-fundamentals.md
■   ■   ■■■ linux-cheatsheet.md
■   ■   ■■■ networking-basics.md
■   ■   ■■■ cryptography.md
■   ■■■ lab-setup/
■   ■   ■■■ setup-steps.md
■   ■■■ screenshots/
■   ■■■ report/
■       ■■■ Task-1-Report.pdf
■■■ Task-2/
■■■ Task-3/
■■■ Task-4/
■■■ Task-5/
```

Before publishing

• Create the repository on GitHub and set it to Public if that is required by the internship instructions.

• Add a README explaining the internship task, learning objectives, lab setup and folder structure.

• Upload notes, command cheat sheets, your final report and sanitized screenshots.

• Never upload passwords, tokens, private keys, personal data, VM disk images or large packet captures containing sensitive data.

• Open the repository in a signed-out/private browser window to verify that the public link works.

8.3 Five-minute LinkedIn video outline

Time	What to explain
0:00–0:30	Introduce yourself and the ApexPlanet Task 1 objective.
0:30–1:20	Explain the CIA Triad and common cybersecurity threats.
1:20–2:20	Show the isolated lab architecture and your Kali/target setup.
2:20–3:20	Demonstrate Linux and networking commands.
3:20–4:20	Show your OpenSSL, Wireshark and authorized lab tool exercises.
4:20–5:00	Summarize what you learned and show your GitHub repository.

Video recording checklist

• Record your screen with readable terminal text and clear narration.

• Show only activities you have actually completed.

• Do not expose passwords, private keys, personal email content or sensitive IP information.

• Upload the video to LinkedIn as instructed and copy the direct video/post URL.

• Check that the video is visible to the intended reviewers before submitting its link.

8.4 ApexPlanet submission form checklist

Form field	What to enter
Full Name	Devaraju GR (as shown in your form)
Email ID	Use the email shown in your ApexPlanet registration.
College Name	Enter your official college name if you want it on the certificate.
Degree Course	Enter your actual degree course, e.g. B.Tech / B.E. only as applicable.
Internship Domain	Cybersecurity & Ethical Hacking (as shown in your form)
Offer Letter ID	Use the ID printed on your own offer letter; verify before submitting.
Explanation Video Task-1	Paste your published LinkedIn video URL.
GitHub Repository Link Task-1	Paste your public Task 1 repository URL.

Submit only after the required practical work, report, repository and video are complete. Save a screenshot or confirmation of the successful submission.

9. Completion tracker

Update this tracker as you complete the activities. The current status is intentionally marked Pending because the practical work has not yet been started.

Activity	Completion evidence	Current status
Cybersecurity basics and CIA Triad	Notes / short explanation	Pending
Threats and attack vectors	Threat table and defenses	Pending
VirtualBox/VMware setup	VM configuration screenshot	Pending
Kali Linux setup	Desktop and terminal screenshot	Pending
Metasploitable 2 / DVWA	Target running in isolated lab	Pending
Linux fundamentals	Commands and permission screenshots	Pending
Networking basics	IP, route and DNS outputs	Pending
Cryptography / OpenSSL	SHA-256 and certificate output	Pending
Wireshark	Own lab packet capture	Pending
Nmap	Authorized lab scan output	Pending
Burp Suite	Local DVWA request screenshot	Pending
Netcat	Local connection demonstration	Pending
GitHub repository	Public repository URL	Not created
5-minute LinkedIn video	Published video URL	Not recorded
ApexPlanet portal submission	Submission confirmation	Not submitted

10. Glossary

Term	Meaning
Asset	Something valuable that needs protection.
Threat	A potential cause of harm to a system or organization.
Vulnerability	A weakness that could be exploited.
Exploit	A method or code that takes advantage of a vulnerability.
Risk	The likelihood and impact of a threat exploiting a vulnerability.
MFA	Multi-factor authentication; more than one authentication factor.
Packet	A formatted unit of data carried over a network.
Port	A logical endpoint used by transport protocols.
Hash	A fixed-length digest calculated from input data.
TLS	Transport Layer Security; protects communication channels.

11. References and learning resources

• ApexPlanet internship task brief supplied by the participant – Task 1: Foundation & Environment Setup.

• Kali Linux documentation: https://www.kali.org/docs/

• Oracle VirtualBox user manual: https://www.virtualbox.org/manual/

• Wireshark User's Guide: https://www.wireshark.org/docs/

• Nmap Reference Guide: https://nmap.org/book/man.html

• OpenSSL documentation: https://docs.openssl.org/

• OWASP Web Security Testing Guide: https://owasp.org/www-project-web-security-testing-guide/

• OWASP Juice Shop / DVWA documentation for intentionally vulnerable local practice applications.

Declaration: This document is a learning report template. Practical completion, screenshots, tool outputs, repository publication and video publication must be supplied by the participant based on work actually performed.

— End of Task 1 starter report —